

Natural Orbitals and Sparsity of Quantum Mutual Information

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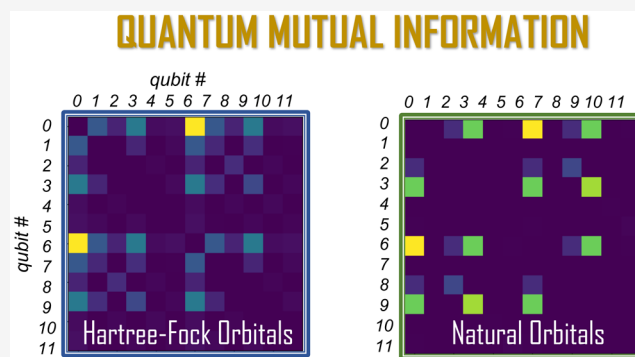


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ABSTRACT: Natural orbitals, defined in electronic structure and quantum chemistry as the molecular orbitals diagonalizing the one-particle reduced density matrix of the ground state, have been conjectured for decades to be the perfect reference orbitals to describe electron correlation. In the present work we applied the Wave function-Adapted Hamiltonian Through Orbital Rotation (WAHTOR) method to study correlated empirical ansätze for quantum computing. In all representative molecules considered, we show that the converged orbitals are coinciding with natural orbitals. Interestingly, the resulting quantum mutual information matrix built on such orbitals is also maximally sparse, providing a clear picture that such orbital choice is indeed able to provide the optimal basis to describe electron correlation. The correlation is therefore encoded in a smaller number of qubit pairs contributing to the quantum mutual information matrix.



1. INTRODUCTION

Electronic structure theory and its implementation into algorithms and computer programs allow the study of many physical and chemical processes with both theoretical and application interests. In cases where electron correlation plays an important role, high-level classical electronic structure and quantum chemistry tools are necessary, albeit they become limited by the bad scaling of the required computational time with the size of the system. Quantum devices may offer new opportunities to tackle these problems, in principle, leading to exponential advantage.^{1–4} Many algorithms have been developed,^{5–9} including hybrid quantum-classical algorithms that use classical and quantum resources in combination, exploiting at best the features of both devices.

Among the hybrid algorithms, the variational ones are widespread in different fields of application such as machine learning,^{10,11} combinatorial optimization,^{12,13} strongly correlated systems,¹⁴ and quantum chemistry.^{15,16} One of the most widely used algorithms is the variational quantum eigensolver (VQE),^{17–19} which has the advantage of being suited for implementation into noisy intermediate-scale quantum (NISQ) devices.^{20–22} The electronic structure problem is tackled by writing the molecular Hamiltonian in the second quantization scheme and mapping Fermions into qubits using one of the many encoding methods available.^{23–25} In this scheme, the Hamiltonian is written as a weighted sum of strings of Pauli operators. Variational parameters of wave function ansätze are therefore optimized using a classical computer, whereas the quantum processor is used to calculate expected values of the Hamiltonian.

Variational wave functions, i.e., variational quantum circuits, used in VQE algorithm are often inspired by existing quantum chemistry methods such as Unitary Coupled Cluster (UCC).^{26,27} These classes of Quantum-Chemistry-inspired ansätze have the advantage of being directly derived by a consolidated theoretical framework, but they have the drawback of being realized by circuits composed by a large number of quantum gates. For instance, in the case of UCCSD the number of gates scales as $O((M - N)^2 N^2)$, where M represents the number of spin orbitals and N the number of electrons.²⁶ On NISQ devices, which are limited by a small number of gates due to the short coherence time of the qubits, heuristic ansätze are rather considered, since they can better exploit the capabilities of the hardware, at the cost of losing the chemical/physical meaning of their construction.²⁸ Many groups have recently developed different variants of the VQE algorithm, with the aim to improve its efficiency, enhance the quality of the wave function, and reduce the circuit depth.^{29–35}

In this article, we analyze the quantum mutual information matrix of shallow depth ansätze in which we optimized both the variational parameters and the orbitals used to construct the ansätze. To achieve this goal, we used a recently optimized version of the Wave function-Adapted Hamiltonian Through Orbital Rotation (WAHTOR) algorithm.^{36,37} The WAHTOR

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algorithm was introduced in order to have a good description of the system considered using a short circuit, improving the energy results without increasing the number of the quantum gates in the circuit. The WAHTOR algorithm finds the optimal unitary transformation of the Hartree–Fock orbitals for a given ansatz of a fixed topology. In this perspective, the Hamiltonian is optimized (or, in another perspective, orbitals are optimized) in order to adapt to the shape of the ansatz. From the computational point of view, this optimization is achieved by using the gradient of the energy with respect to the parameters of the unitary transformation, without significant overload. In an improved version of the algorithm,³⁷ we exploited higher-order derivatives, reformulating the algorithm for a more efficient and faster convergence.

By analyzing the resulting optimized orbitals, we have observed that they are converging very closely to the natural orbitals basis,^{38–40} namely, the orbitals determined by the unitary transformation that diagonalizes the one-particle reduced density matrix of that ground state. The WAHTOR procedure seems therefore to lead to the finding of “maximally correlated orbitals” which were indeed identified as natural orbitals through the years. In addition, anticipating our results, we have discovered that in this WAHTOR-optimized/natural orbital basis the sparsity of the quantum mutual information matrix is indeed minimal, as expected when the basis set better expresses the electronic correlation of the system.

In section 2 we shortly review the improved WAHTOR algorithm, and then we show the computational details of the algorithm, the systems studied, the ansätze used, and the analysis performed. Section 3 shows how, for the systems of study, the optimized basis sets converge to these of natural orbitals. Finally, in the last section we focus on the key results obtained and related discussions.

2. COMPUTATIONAL DETAILS

WAHTOR Algorithm. The main idea of the WAHTOR algorithm is to adapt the Hamiltonian to a given empirical quantum circuit.³⁶ In this regard we exploit the invariance of the Hamiltonian eigenvalues with respect to unitary transformations. In particular, the Hamiltonian, in the second quantization formalism, is the following:

$$H = \sum_{i,j} h_{ij} a_i^\dagger a_j + \frac{1}{2} \sum_{c,d,e,f} g_{cdef} a_c^\dagger a_d^\dagger a_e a_f \quad (1)$$

where the sums run on M , the number of spin orbitals, and h_{ij} and g_{cdef} are respectively the one-body and two-body integrals. We choose the following particular form for the unitary operator $\hat{U}$:

$$\hat{U} = \hat{U}(\mathbf{R}) = e^{\sum_k (i\mathbf{R} \cdot \mathbf{T})_k a_k^\dagger a_k} \quad (2)$$

where the components of $\mathbf{T}$ are the generators of the unitary Lie group with dimension $M \times M$, where M is the number of spin orbitals. The elements of the group are obtained by linearly combining the generators with real coefficients, represented by the components of the vector $\mathbf{R}$. The operator $\hat{U}^\dagger \hat{H} \hat{U}$ is the Hamiltonian H expressed in a new basis set. Given a parametrized quantum state $|\Psi(\boldsymbol{\theta})\rangle$ on a quantum circuit, we can define the energy cost function

$$E(\mathbf{R}, \boldsymbol{\theta}) = \langle \Psi(\boldsymbol{\theta}) | H(\mathbf{R}) | \Psi(\boldsymbol{\theta}) \rangle \quad (3)$$

Both sets of parameters can be alternatively optimized: the $\boldsymbol{\theta}$ parameters with a VQE algorithm and the $\mathbf{R}$ parameters with one of the procedures detailed in the nonadiabatic version of the WAHTOR algorithm,³⁷ which is the method used in this work. When convergence is reached, we obtain the converged set of spin orbitals.

Simulated Systems. We considered seven different molecular systems: H_2 , LiH , HF , BeH_2 , H_2O , H_2S , and NH_3 . Each system was considered at the equilibrium geometry, and bond distances are reported in the Supporting Information. Depending on the systems, we employed various basis sets, as specified in Table 1. Frozen core approximation apply to all molecules except H_2 system.

Table 1. Basis Set, Number of Qubits, and Groups of Orbitals with Same Geometrical Symmetries for Each Molecule under Investigation

molecule	basis set	qubits	orbitals groups	# of generators
H_2	6-31G	8	{1,3},{2,4}	2
LiH	STO-3G	10	{1,2,5}	3
HF	STO-3G	10	{1,2,5}	3
H_2	6-311G*	12	{1,3,5},{2,4,6}	6
BeH_2	STO-3G	12	{1,5},{2,6}	2
H_2O	STO-3G	12	{1,3,5},{2,6}	4
H_2S	STO-3G	12	{1,3,6},{2,5}	4
NH_3	STO-3G	14	{1,4,5},{2,6},{3,7}	5

We used the Jordan–Wigner encoding method,²³ obtaining a number of qubits ranging from 8 to 14.

The Hartree–Fock orbitals and the terms of the molecular Hamiltonian have been obtained using the PySCF package.⁴¹ The WAHTOR algorithm has been implemented by a in-house produced Python program exploiting the Qiskit libraries.⁴² Both the VQE and the WAHTOR algorithm were performed for each molecule, and the energy results obtained were compared taking into account the fraction of correlation energy ϵ , defined by

$$\epsilon = \frac{E_{\text{calculated}} - E_{\text{HF}}}{E_{\text{full-CI}} - E_{\text{HF}}} \quad (4)$$

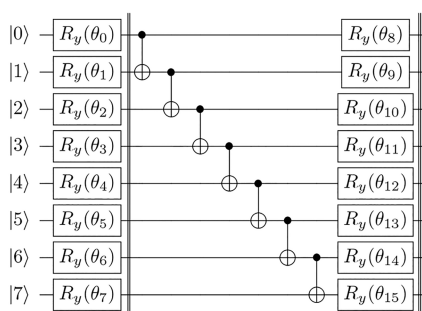
where $E_{\text{full-CI}}$ is the exact energy, $E_{\text{calculated}}$ is the energy calculated using VQE or WAHTOR algorithm, and E_{HF} is the Hartree–Fock energy. The percentage of correlation energy values are shown in Table 2 for each molecule under investigation.

Implementation of Quantum Algorithm. The parametrized quantum circuit chosen in the present work belongs to the so-called heuristic ansatzes, and it is built as follows. First, a repetition of single qubit rotations around the y -axis is applied to the n -qubits Hartree–Fock reference state. After such rotations, an entangling block composed of CNOT gates followed by another set of independent rotations on each qubit has been applied. The block composed by CNOT gates and rotations can therefore be repeated a certain number of times, each time using different variational parameters, as shown in Figure 1. The total number of blocks determines the circuit depth. In this work, we considered depth 2 for 8 and 10 qubits systems, depth 4 for 12 qubits systems and depth 6 for the NH_3 molecule. In all systems the CNOT gates are arranged to form a ladder, as shown in Figure 1.

For each molecular system, we executed 100 VQE runs starting from random variational parameters, and the lowest

Table 2. VQE and WAHTOR Percentage Correlation Energy Results and Geometric Distance from the Reference Natural Orbitals for Each Molecule under Investigation

molecule	FCI (E_h)	VQE (E_h)	% VQE	WAHTOR (E_h)	% WAHTOR	δ
H ₂	−1.151673	−1.133047	25.25	−1.146234	78.17	0.997
LiH	−7.882174	−7.876560	72.14	−7.881116	94.75	0.999
HF	−98.596603	−98.591465	80.11	−98.594804	93.03	0.999
H ₂	−1.153472	−1.139391	44.74	−1.148713	81.32	0.860
BeH ₂	−15.594691	−15.575995	46.36	−15.576002	46.38	HF⟩ ≈ NO⟩
BeH ₂	−15.594691	−15.568763	25.61	−15.569138	26.68	HF⟩ ≈ NO⟩
BeH ₂	−15.594691	−15.566201	18.25	−15.566222	18.31	HF⟩ ≈ NO⟩
H ₂ O	−75.012359	−74.988750	52.22	−74.999297	73.56	0.971
H ₂ S	−394.353959	−394.332052	48.34	−394.336006	57.66	0.888
NH ₃	−55.520291	−55.480073	38.87	−55.480277	39.18	0.999

**Figure 1.** Ladder entangler map for an 8-qubits system. The quantum gates between the two barriers compose the block (identified by bold lines in the figure). The block is repeated a certain number of times d , which defines the depth of the circuit.

energy result was considered as the starting point for the WAHTOR algorithm. We optimized the ansatz parameters using the L-BFGS-B optimizer,⁴³ whereas the Hamiltonian optimization step is performed by applying the nonadiabatic trust region optimizer defined in ref 37. The convergence is reached when two successive VQE energies in the WAHTOR algorithm differ by less than 10^{-6} Hartree. All simulations are executed using the statevector simulator.

Hamiltonian Parameters. The implementation of the WAHTOR algorithm for a system consisting of m Fermionic modes orbitals is realized by the unitary matrices that determine the basis change. These matrices are developed starting from a set of m^2 generators, comprised of $\frac{m^2+m}{2}$ symmetric and $\frac{m^2-m}{2}$ antisymmetric. The number of Hamiltonian parameters is in principle equal to the number of generators, but in practice, it can be significantly reduced, without losing its accuracy and increasing the efficiency of the algorithm. To this purpose, we take into account spin symmetry, ensuring that spin-up and spin-down orbitals are never linearly combined during optimization. Therefore, we can exclude the generators mixing spin up and spin down. In addition, given that the Hamiltonians are real, our attention is directed solely toward employing generators that yield real unitary matrices, we are considering transformations belonging to the orthogonal group. Furthermore, we constrain the optimization process to linearly combine only orbitals with the same molecular symmetries. The list of groups of orbitals sharing the same symmetries is provided, for each molecule, in the fourth column of Table 1. These constraints streamline the problem, reducing the number of generators, i.e., the components of the vector T , as well as the parameters R to

be optimized. Consequently, this reduction leads to a decrease in the overall computational cost. The number of generators for each system is shown in the final column of Table 1. To clarify our procedure, as an example, we consider the H₂O molecule in the STO-3G basis set, mapped on a register of 12 qubits. It follows that 144 generators are required to implement a generic unitary transformation. By taking into account the spin symmetry, the number of generators is reduced to 36. Since the Hamiltonian is real, we just consider transformations belonging to the orthogonal group, reducing to 15 the number of generators. Finally, we only linearly combine orbitals with the same geometric symmetries. For this system, we obtain 2 sets of orbitals with the same geometric symmetries containing more than one element. The indices of the orbitals belonging to each set is shown in Table 1. As a consequence, the final number of generators is only 4 (3 generators for the first group and 1 for the second group).

Quantum Mutual Information Matrix. Since we use the Jordan-Wigner encoding method, correlations between qubits is reflected into the correlation between the occupation numbers of different spin orbitals encoded into the corresponding qubits. A useful measure of correlation between the qubit pairs is represented by the quantum mutual information.⁴⁴ Considering a system of n qubits, each one defined in its own Hilbert space $\mathcal{H}_i$ for $i = 1, \dots, n$, the state of the composite system $|\Psi\rangle$ belongs to the space composed by the tensor product of the individual spaces $|\Psi\rangle \in \bigotimes_{i=1}^n \mathcal{H}_i$. The density operator has the following expression:

$$\rho = |\Psi\rangle\langle\Psi| \quad (5)$$

By defining the set of indices $\Omega = \{1, \dots, n\}$ we can select a set of indices $k \in \Omega$. The reduced density matrix on qubits with indices in k is obtained from the density matrix ρ tracing out over all the indices with the exception of k :

$$\rho_k = \text{Tr}_{\{\Omega-k\}}(\rho) \quad (6)$$

In particular, we define the one-qubit and two-qubits reduced density matrices as

$$\rho_i = \text{Tr}_{\{\Omega-i\}}(\rho) \quad (7)$$

$$\rho_{ij} = \text{Tr}_{\{\Omega-(i,j)\}}(\rho) \quad (8)$$

where i and j refer to the qubits considered. Now, the calculation of Von Neumann entropy S

$$S(\rho) = -\text{Tr}(\rho \ln \rho) \quad (9)$$

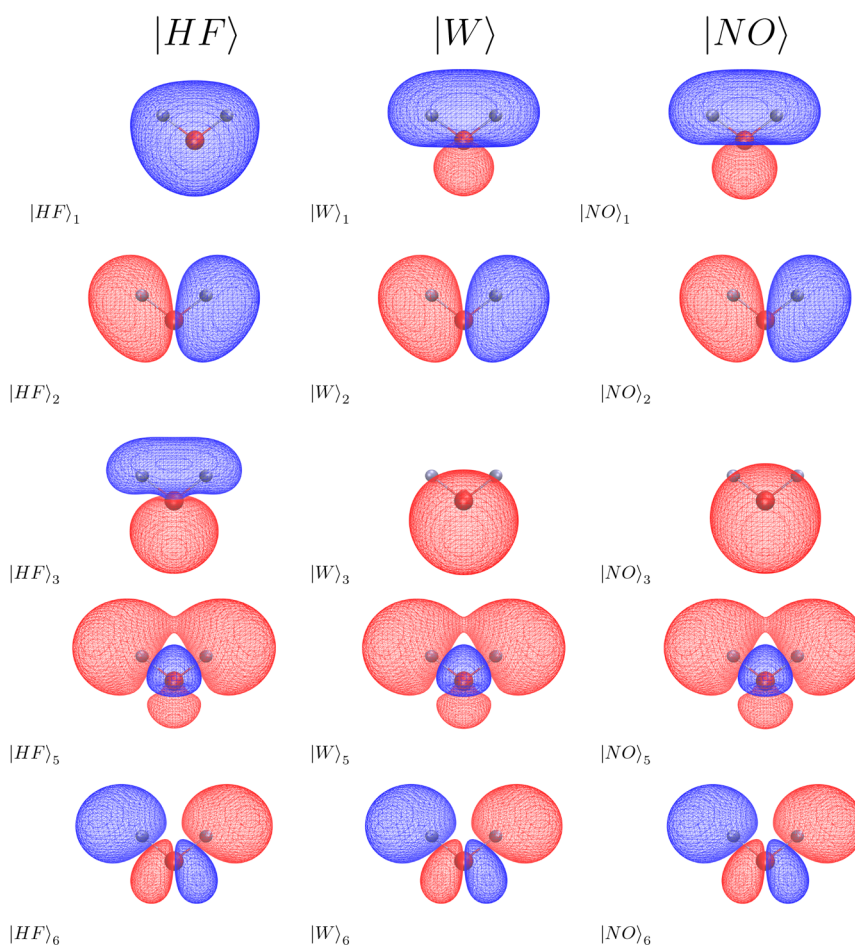


Figure 2. Representation of the H_2O molecular orbitals in the frozen core approximation optimized by the algorithm. Each column corresponds to a different basis set: the first one represents the Hartree–Fock orbitals, the second one the WAHTOR-optimized orbitals, and the one the natural orbitals.

for the reduced density matrices defined in eqs 7 and 8 can be obtained as follows:

$$S(\rho_i) = -\text{Tr}(\rho_i \ln \rho_i) \quad (10)$$

$$S(\rho_{ij}) = -\text{Tr}(\rho_{ij} \ln \rho_{ij}) \quad (11)$$

Finally, the quantum mutual information between qubits i and j is defined as

$$I_{ij} = (S(\rho_i) + S(\rho_j) - S(\rho_{ij}))(1 - \delta_{ij}) \quad (12)$$

We represent the quantum mutual information as a symmetric matrix with elements along the diagonal equal to zero by definition, following ref 34. The value of the correlation between the qubits can be observed in the upper or the lower triangular part of this matrix.

Orbital Analysis. In order to carry out an analysis on the convergence of the orbitals obtained with the WAHTOR algorithm to the natural ones, as conjectured, the dimensionless parameter $\delta \in [0, 1]$ has been defined. It measures how much the converged WAHTOR orbitals ($|W\rangle$) are close to the natural ones ($|NO\rangle$), calculated from the exact Full-CI state.⁴⁰ The definition of this parameter is the following,

$$\delta = \frac{\sum_i (|\langle W_i | NO_i \rangle| - |\langle HF_i | NO_i \rangle|)}{N - \sum_i |\langle HF_i | NO_i \rangle|} \quad (13)$$

where N is the number of orbitals considered and $|HF\rangle$ are the Hartree–Fock ones.^{45,46} So the closer the δ value is to 1, the more the optimized orbitals are similar to the reference natural orbitals (corresponding to $\delta = 1$); the closer it is to zero, the more the optimized orbitals are closer to the Hartree–Fock orbitals (corresponding to $\delta = 0$).

3. RESULTS

In this section we show the results obtained for the molecules taken into consideration: for each studied system the energy results obtained with the VQE and the WAHTOR algorithms have been compared, as well as the convergence of the single-particle basis set to the natural orbitals. The results are summarized in Table 2: the first column of the table reports the molecule under consideration, while the second column is the reference energy expressed in Hartree (E_h) calculated using the Full Configuration Interaction (FCI) method. In the third and fourth columns, the lowest energy values obtained from 100 VQE runs and the related percentage correlation energies are reported, respectively. The fifth and sixth columns show the converged energies of the WAHTOR algorithm and the corresponding percentages of correlation energies. We note that, for each molecule, our method always improves the VQE results. Interestingly, in the case of BeH_2 the orbital optimization does not change significantly the energy value for all of the three VQE results considered; the reasons for this

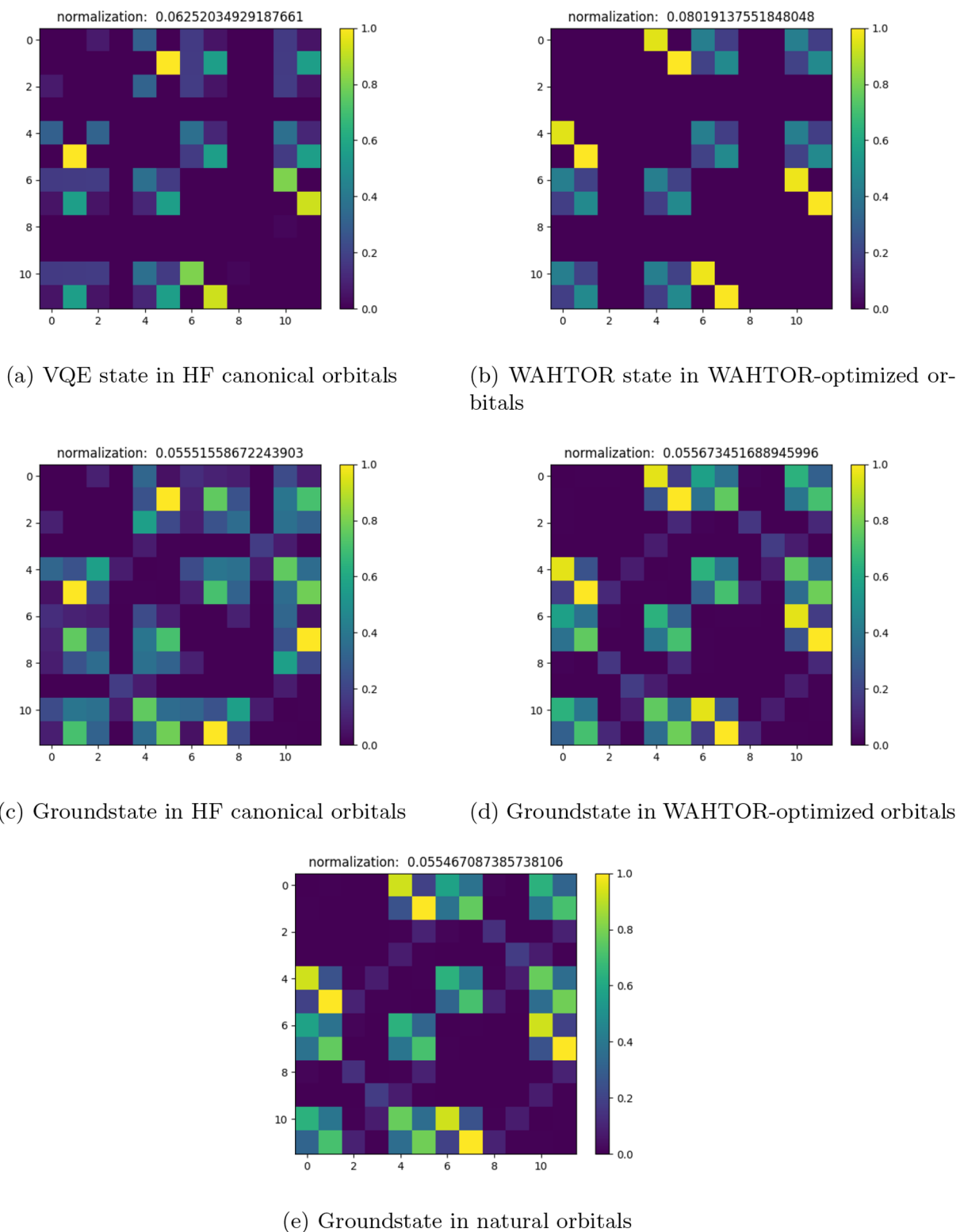


Figure 3. Quantum mutual information maps for H_2O molecule. States and basis sets considered are described in the captions.

pitfall will be clearer after the orbital analysis, and we will come back to this point later in the text. In the last column of Table 2 we reported the δ parameter, as defined by eq 13 in the computational details. This parameter is a measure of the distance between optimized orbitals and natural orbitals, ranging from $\delta = 0$ (Hartree–Fock orbitals) to $\delta = 1$ (natural orbitals of the ground state calculated with FCI). Focusing on

the H_2O molecule, we note that $\delta = 0.971$, so the optimized single-particle basis set corresponds approximately to the natural orbitals. The similarity is evident as it can also be shown by comparing HF canonical orbitals, WAHTOR-optimized orbitals, and natural orbitals reported respectively in the first, second, and third columns of Figure 2. Each row in the figure shows the five orbitals that are optimized by our

algorithm. The orbitals in the first column have been obtained with the restricted Hartree–Fock method using the frozen core approximation. We split the orbitals into two groups that contain the ones with the same geometric symmetries: $|HF\rangle_1$, $|HF\rangle_3$ and $|HF\rangle_5$ belong to the first group, while $|HF\rangle_2$ and $|HF\rangle_6$ belong to the second one. As explained in section 2, two orbitals are combined linearly only if they belong to the same group so, since the missing orbital in Figure 2 has a unique geometric symmetry, it has not been optimized by the algorithm and therefore has not been reported. The WAHTOR-optimized orbitals in the second column of Figure 2 show that only the orbitals $|HF\rangle_1$ and $|HF\rangle_3$ have been appreciably modified, becoming $|W\rangle_1$ and $|W\rangle_3$. By comparing the second and third columns of the figure, we infer that the WAHTOR-optimized and natural orbitals are almost identical. The same conclusions can be inferred by taking into account the other molecular systems reported in Table 2. Indeed, also in these cases, the δ parameter is close to 1, almost independently by the energy reached by the optimization algorithm. An exception is made by the BeH_2 molecule for which the WAHTOR algorithm does not optimize the VQE energies, as shown in the last column of Table 2. To emphasize this result, we considered the best three different VQE energies among 100 runs and tried to optimize the cost function without significant improvements. In terms of our previous statements, this is not surprising if we consider that the natural orbitals and the Hartree–Fock orbitals are almost identical for this system. We have chosen more than one independent VQE solution to demonstrate that the missing optimization does not depend on the starting wave function.

In summary, our WAHTOR procedure is leading to orbitals which are extremely close to natural orbitals. We can therefore make the conjecture that our algorithm is guiding the Hamiltonian (i.e., rotating the orbitals) in such a way that they are getting closer to the natural orbitals of the groundstate expressed in the Hartree–Fock basis set, which in principle can be obtained only by diagonalizing the one-body density matrix of this state.

In order to better understand and interpret the obtained result, let us proceed with a deeper analysis of the correlation between the occupation number of the spin orbitals of the molecules. In the following, we will investigate the H_2O molecule, albeit the same conclusions can be found when analyzing the other systems, as illustrated in the Supporting Information. We would like first to point out that, by using the Jordan–Wigner mapping, the occupancy of each spin orbital is associated with the state of a single qubit. Therefore, on a given quantum state, the correlations between the occupancy of the spin orbitals can be analyzed through the study of the correlations between the qubits, and for this purpose, we use the quantum mutual information matrix defined in the previous section. In particular, Figure 3 shows the quantum mutual information matrices for the states of the H_2O molecule. Considering that the first half of the qubits represent the orbitals with spin up while the second half represents the same orbitals with spin down, each matrix is symmetrical and we can focus only on the lower or upper triangular part. We define the VQE and WAHTOR states as the ones obtained after the application of the corresponding algorithm. Then the VQE state is expressed in the Hartree–Fock molecular basis set and is shown in panel (a) of Figure 3, while the WAHTOR state, expressed in the WAHTOR-optimized single-particle basis set, is reported in panel (b). Panels (c), (d), and (e)

show the quantum mutual information matrices for the ground state expressed in the Hartree–Fock, WAHTOR-optimized, and natural orbitals, respectively. The colored spot ranges from 0 to 1 with the normalization constant reported above each figure. First of all, we note that the maximal value of the correlation for image (b) is 33% higher than the one in image (a). Moreover, the number of green and yellow colored spots in panel (b) is increased with respect to the ones in panel (a), while the number of blue spots decreases. This means that the quantum mutual information, during the optimization, has been compacted on a lower number of qubits pairs so that the sparsity of the quantum mutual information matrices is increased for the states expressed in the WAHTOR-optimized orbitals. This feature can be observed also comparing the ground state expressed in Hartree–Fock and WAHTOR-optimized molecular basis set reported in panels (c) and (d), respectively. Finally, we note that figures (d) and (e) are identical, confirming in terms of the correlation between the occupation number of spin orbitals that the single-particle basis set obtained with the optimization process is almost identical to the natural one.

Figure 4 quantifies the variations in quantum mutual information between qubits pairs sorted in a decreasing way

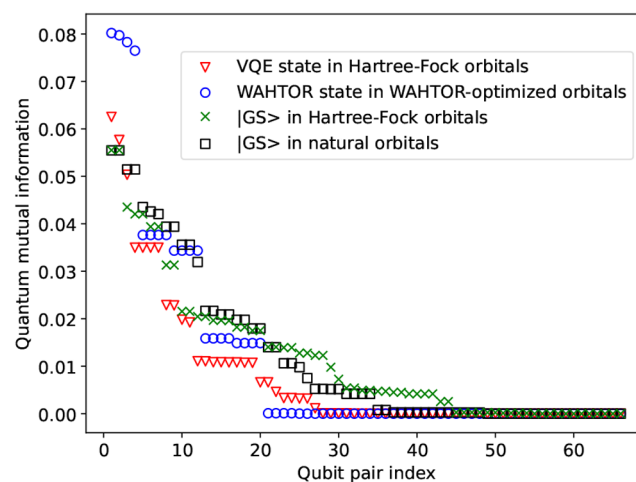


Figure 4. Quantum mutual information values for qubit pairs in the H_2O molecule are shown ordered in descending order. For the exact ground state results (green cross and black squares) is evident how the quantum mutual information is concentrated on a smaller number of qubit pairs in the natural orbital basis. The same finding is achieved for the variational solutions (red triangle and blue circles). In the case of WAHTOR-optimized orbitals only about 20 qubit pairs have quantum mutual information values significantly different from zero.

for both the exact and variational wave functions represented in the matrices of Figure 3a–d. We note that for the exact ground state the number of qubit pairs significantly different from zero is smaller when the wave function is expressed in natural orbitals with respect to the case of Hartree–Fock orbitals. The same trend is observed for the variational wave functions, confirming the reduction of the sparsity in the matrices illustrated in Figure 3. Moreover, the maximal value of correlation between the qubit pairs is increased during the optimization passing from the state of Figure 3a to the state of Figure 3b. In the Supporting Information, we show that there is a reduction of the sparsity in the matrices also when the other molecules shown in Table 2 are taken into account.

Focusing on the converged molecular basis set, it has been conjectured in ref 38 that, for a given accuracy, the natural orbitals are the single-particle basis set in which the wave function is expressed with the lowest number of Slater determinants. In terms of correlations between qubits, we expect that as the number of determinants in the wave function increases, the more the number of colored spots in the quantum mutual information matrices increases. This interpretation explains why the sparsity of the quantum mutual information matrices is increased when we express the states in the natural orbitals basis set. To further check the convergence of the WAHTOR procedure to natural orbital, we decided to run VQE simulations for the H_2O molecule directly starting from the natural orbitals. As expected, the VQE lowest energies reached the 73% of correlation energy, as found at the end of WAHTOR optimization. Additional details can be found in the [Supporting Information](#).

4. CONCLUSIONS

Natural orbitals diagonalize the one-particle reduced density matrix, and it is supposed they are the orbitals in which the ground state is represented by a minimal number of Slater determinants. In this work we have investigated whether our orbital optimization procedure of an empirical ansatz for quantum chemistry is driving orbitals toward natural orbitals. For this purpose we have introduced a parameter δ and, as reported in [Table 2](#), we show that it is very close to 1 in the case of optimized WAHTOR orbitals for different benchmark molecules. This convergence to natural orbitals is reached regardless of the geometric symmetries, the number of qubits, and the accuracy achieved from the VQE results, for the molecules under consideration.

This effect is also reflected in terms of correlations: the comparison of the quantum mutual information matrices built on Hartree–Fock and natural orbitals demonstrates that in the natural orbital basis set there is an increase of the sparsity of the matrix, as reported in [Figure 3](#) for the case of the H_2O molecule. Similar considerations can be deduced for all other molecules by comparing the pictures in the [Supporting Information](#). The identification of natural orbitals as optimal orbitals for the building of empirical ansätze may suggest them as being the best candidate one-electron basis for developing compact empirical wave function for state preparation on NISQ devices.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.3c01325>.

Molecular geometries, quantum mutual information, and distribution of the energy results ([PDF](#))

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Notes

The authors declare no competing financial interest.

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